Native Galois fields of small wild cycles

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Abstract

For every odd prime p, we determine the Galois action on all canonical small cycles of $P(z) = (1+s)z + z^2$ over $K = \overline{\mathbb{F}}_p((s))$. The degree-$p^e$ small factor is irreducible for every $e \geq 1$; each root generates a totally ramified cyclic splitting field of degree p^e, whose distinguished generator is one application of P. The common root valuation has denominator only p, so it does not establish this degree. We instead prove an exact second-layer contact formula and transfer the native action through a canonical matching of the top clusters. The degree-p Artin–Schreier classes, oriented by translation by $+1$, agree at every level $e \geq 2$, whereas the prime-period field intersects every later field trivially over K. We compute the common first quotient break $2(p-1)$ and the two upper breaks $2(p-1), 2p(p-1)$ of the second-layer field. Finally, using the independently proved compact adding-machine limit of the cycles, we obtain exact eventual agreement of every fixed finite native Galois quotient and a canonical $\mathbb{Z}_p$-extension. This last tower consists of stabilized quotient fields, not the original sequence of full periodic-point fields.

1 Introduction

A unique geometric periodic cycle need not be one Galois orbit over the coefficient field. This distinction is particularly pronounced for wild cycles: a single cycle of length p^e permits a proper Galois subgroup of its native rotation group, even when every point has nonintegral valuation. We study this distinction for the marked quadratic multiplier deformation

$$k = \overline{\mathbb{F}_p}, \qquad R = k[[s]], \qquad K = k((s)), \qquad P(z) = (1+s)z + z^2,$$

where p is an odd prime and $v(s) = 1$. All algebraic fields below lie in one fixed separable closure K^{sep}. The word *native* means that the distinguished action is one application of P.

For $e \geq 1$, put

$$Q_e(z) = \frac{P^{\circ p^e}(z) - z}{P^{\circ p^{e-1}}(z) - z}. \tag{1.1}$$

Here and throughout, the superscript $\circ$ denotes composition. The canonical small factor $M_e \in R[z]$ is the monic degree-p^e Hensel factor of Q_e reducing to z^{p^e}. Its roots, denoted S_e, form the unique ordinary p^e-cycle in the open unit disk. These geometric facts follow from the optimal-cycle theorem of Lindahl–Rivera-Letelier [7, Theorem C and Proposition 4.4]; we give the precise factor interface in Section 2.

Theorem 1.1 (Native fields and oriented first quotients)**.** *For every odd prime p and every $e \geq 1$, M_e is irreducible over K. For any $\alpha_e \in S_e$, the field $L_e = K(\alpha_e) = K(S_e)$ is totally ramified cyclic of degree p^e. It has a distinguished generator $\sigma_e(\alpha_e) = P(\alpha_e)$.*

Let F_e be the unique degree-p subfield of L_e. Choose $y_e \in F_e$ with $\sigma_e(y_e) - y_e = 1$, and set $a_e = y_e^p - y_e \in K$. Then, for every $e \geq 2$,

$$[a_e] = [a_2] \text{ in } K/\wp(K), \qquad F_e = F_2, \qquad L_1 \cap L_e = K, \quad \wp(b) = b^p - b. \tag{1.2}$$

The equalities concern oriented classes and embedded fields.

The choice of y_e does not affect its class, once the translation by $+1$ is fixed. Section 5 supplies a trace resolvent with exactly this convention. Equality in (1.2) does not assert equality of its raw Laurent-series representatives.

Theorem 1.2 (First quotient and second-layer ramification)**.** *The unique break of L_1/K is $p-1$. For every $e \geq 2$, the unique break of F_e/K is $2(p-1)$. The upper and lower breaks of L_2/K are respectively*

$$(u_1, u_2) = (2(p-1),\, 2p(p-1)), \qquad (b_1, b_2) = (2(p-1),\, 2(p-1)(p^2-p+1)).$$

The field different exponent, measured by $v_{L_2} = p^2 v$, is

$$d_{L_2/K} = (p-1)(2p^3 - 2p^2 + 3p - 1). \tag{1.3}$$

The degree proof has two distinct steps. First, the contact with the prime-period cycle has exact denominator p^2:

$$v(\alpha_e - \beta_1) = \frac{2(p-1)^2}{p^2} \qquad (e \geq 2).$$

A compositum ramification argument forces degree at least p^2. This contact does not acquire denominator p^e at higher levels. Instead, the multiplier of the level-two cycle gives a cross-level contact deeper than the separation of the p top clusters. The resulting canonical matching transfers a nontrivial native quotient action and forces the entire cyclic Galois group. Its native equivariance also proves the oriented class equality. For the precise ramification invariants, we use a controlled cancellation in L_2L_1, where the relevant element has valuation divisible by p, rather than applying a prime-to-p leading-term rule outside its hypotheses.

Eventual quotients and their additional dependency

Write $\rho_e : G_K \to \mathbb{Z}/p^e\mathbb{Z}$ for the native rotation character, where $G_K = \mathrm{Gal}(K^{\mathrm{sep}}/K)$. A separate consequence, proved in Section 7, is the following.

Theorem 1.3 (Eventual native quotient tower)**.** *For every odd prime p, there is a unique continuous surjection $\chi_\infty : G_K \to \mathbb{Z}_p$ such that*

$$\forall j \geq 1\ \exists E_j \geq j\ \forall e \geq E_j\ \forall g \in G_K : \quad \rho_e(g) \bmod p^j = \chi_\infty(g) \bmod p^j. \tag{1.4}$$

The fields

$$K_j = (K^{\mathrm{sep}})^{\ker(\chi_\infty \bmod p^j)}$$

are nested cyclic degree-p^j extensions of K. For $e \geq E_j$, K_j is the unique degree-p^j subfield of L_e. Their union has Galois group $\mathbb{Z}_p$ over K.

Only Theorem 1.3 uses the compact adding-machine limit proved in the unpublished companion [1, Theorem 1.1]. We state its exact compactness, full-sequence Hausdorff convergence and native conjugacy interface before using it. That theorem is proved independently of Theorem 1.1; conversely, the proofs of Theorems 1.1 and 1.2 use no compact-limit input. In particular, (1.4) implies neither $E_j = j$ nor $K_j = L_j$, and it does not make the original L_e a nested sequence.

Relation to earlier work

Lindahl–Rivera-Letelier own the existence, uniqueness and optimal radius of the small quadratic cycles [7]. The common valuation is $(p-1)/p$, with denominator only p, independently of e. It therefore does not by itself prove degree p^e over K. The arithmetic step here is the uniform second-layer obstruction and the cross-level cluster transfer.

Keating's study of periodic-point fields [5, Theorem 6.1] is a close antecedent: its proof uses commuting Galois actions and a full valuation denominator to establish the degree of a periodic-point extension. Its hypotheses include a finite extension of $\mathbb{Q}_p$, $p > 3$, a ramification bound and a tame base change. His equal-characteristic Proposition 5.1 is also relevant, but compares already cyclic extensions by an isomorphism that may induce a nontrivial automorphism of the base. Neither formulation is an equality of our prescribed characters over the unchanged $k((s))$.

Compatible tower construction through norm fields is also classical; Keating's equivalence [6, Theorem 1.1 and Section 3] allows the base field to vary and works with finite residue field. The distinction in Theorem 1.3 is eventual equality for every fixed quotient of the *prescribed* cycle sequence. The extraction of Galois characters from dynamical torsors has further antecedents; for instance, Debaisieux [3, Propositions 2.1–2.3] uses consistent root sequences for a mixed-characteristic commuting pair. We do not claim the general torsor-to-character step as new. Artin–Schreier theory, ramification filtrations and their quotient compatibility are classical throughout [4].

Our conclusions are local at a fixed multiplier parameter. They do not classify global dynatomic components, compute every higher Witt coordinate, or determine all intersections of full higher periodic-point fields. The comparison with earlier work is bounded to the cited results and makes no universal priority claim.

2 Canonical factors and local ramification tools

Fix an odd prime p, and retain R, K, P from the introduction. The valuation v extends uniquely to an algebraic closure of the complete field K. Write $r = (p-1)/p$.

Lemma 2.1 (Small-cycle factor)**.** *For each $e \geq 1$, there is a unique coprime Hensel factorization*

$$Q_e = M_e V_e, \qquad \overline{M}_e = z^{p^e}, \qquad \overline{V}_e(0) \neq 0,$$

with both factors monic in $R[z]$. The roots $S_e \subset K^{\mathrm{sep}}$ of M_e are p^e distinct points, form one ordinary native p^e-cycle, and all have valuation r. If $v(x) > 0$, then $v(V_e(x)) = 0$.

Proof. We specify the geometric input from [7, Theorem C, its $q = 1$ case, and Proposition 4.4]. For $g(z) = z + z^2$ in odd characteristic it gives

$$\operatorname{ord}_z(g^{\circ p^j}(z) - z) = 1 + \frac{p^{j+1} - 1}{p - 1} \qquad (j \geq 0),$$

and for P it gives the unique cycle of p^e distinct points in the open unit disk, with ordinary least period p^e. It applies after passing to a complete algebraic closure; the roots below are already separable algebraic over K.

For any polynomial h, $h(z) - z$ divides $h^{\circ p}(z) - z$: in the quotient by $h(z) - z$, every iterate of h equals z. Monic division over R thus proves $Q_e \in R[z]$. Reducing and subtracting the two displayed orders gives $\overline{Q}_e = z^{p^e} \overline{V}_e$, where $\overline{V}_e(0) \neq 0$. Coprime Hensel factorization gives the unique factors. Their coefficients are integral, so V_e evaluates to a unit at every point of positive valuation. Every point of the cited cycle is therefore a root of M_e; there are exactly $\deg M_e = p^e$ such distinct points. This proves both separability and identification of its roots.

For completeness, their common valuation can be recovered from the factor. On points of positive valuation, $P(x) = x(1+s+x)$ has the same valuation as x. Since the roots form one cycle, their valuations coincide. Differentiating the numerator and denominator of (1.1) at the common root 0, or cancelling their simple factors z, gives

$$Q_e(0) = \frac{(1+s)^{p^e} - 1}{(1+s)^{p^{e-1}} - 1} = s^{p^e - p^{e-1}}.$$

The denominator is nonzero. Since $V_e(0)$ is a unit, the product of all p^e roots has valuation $p^e - p^{e-1}$. Their common valuation is r. □

Lemma 2.2 (Rotation subgroup)**.** *For $\alpha \in S_e$, $K(\alpha)$ is the splitting field of M_e, with Galois group a subgroup $H_e \leq \mathbb{Z}/p^e\mathbb{Z}$ of native rotations. Its degree is p^{h_e}, where $1 \leq h_e \leq e$, and it is totally ramified. If $h_e < e$, every nonidentity automorphism has native rotation index divisible by p. At $e = 1$ the degree is p.*

Proof. Every root is an iterate of α, so lies in $K(\alpha)$. The factor is separable by Lemma 2.1, hence this root field is Galois. Galois commutes with P. A permutation commuting with a single cycle is determined by the image of one point and is a rotation of that cycle. Thus the splitting group is a subgroup of the cyclic group of order p^e. The valuation $v(\alpha) = r$ forces its ramification degree, and hence its degree, to be divisible by p. Every finite extension of K has trivial residue extension because k is algebraically closed; the degree–ramification identity for complete discretely valued fields makes it totally ramified. A proper subgroup of $\mathbb{Z}/p^e\mathbb{Z}$ lies in $p\mathbb{Z}/p^e\mathbb{Z}$. At $e = 1$ the lower and upper degree bounds agree. □

Lemma 2.3 (Native displacement)**.** *For points x, y with $v(x), v(y) \geq r$, the map P is an isometry and, for $a \geq 0$ and $x \neq y$,*

$$\frac{P^{\circ a}(x) - P^{\circ a}(y)}{x - y} \in 1 + \{u : v(u) \geq r\}. \tag{2.1}$$

For $x \in S_e$, a step of index prime to p has displacement valuation $2r$; every step of index divisible by p has displacement valuation at least $3r$, with $v(0) = +\infty$.

Proof. The disk $v(z) \geq r$ is P-invariant, and

$$P(x) - P(y) = (x - y)(1 + s + x + y).$$

The second factor belongs to the set in (2.1); products of such factors remain in that set. This proves the first assertion by iteration. For $x \in S_e$,

$$v(P(x) - x) = v(sx + x^2) = 2r$$

because $1 + r > 2r$. In the telescoping sum of a one-step displacements, each summand divided by the first has residue 1, by (2.1). If $p \nmid a$, their sum has nonzero residue a, giving valuation $2r$. For $a = p$, the residues cancel in characteristic p, and every remaining term gains at least r. Thus the p-step displacement has valuation at least $3r$. Telescoping in blocks of p proves the last assertion. □

We use lower and upper ramification groups G_t, G^u with the following convention. For a totally ramified finite Galois extension D/K, put $v_D = [D : K]v$, and for $g \neq 1$ set

$$\ell_D(g) = v_D(g\pi - \pi) - 1, \qquad \psi_D(u) = \int_0^u [G : G^t]\, dt,$$

where π is a uniformizer and $G = \mathrm{Gal}(D/K)$. The first lower and upper breaks coincide. Upper numbering commutes with Galois quotients. Abelian upper breaks are integral; in a cyclic degree-p^2 extension,

$$b_2 - b_1 = p(u_2 - u_1), \qquad u_2 \geq pu_1. \tag{2.2}$$

A nontrivial degree-p character over K has a reduced Artin–Schreier polar representative whose largest pole is prime to p and equals its break. These classical facts hold for the arbitrary perfect residue field k, not just finite residue fields [4, Section 2].

Lemma 2.4 (Prime-to-p leading value)**.** *Let D/K be a finite totally ramified Galois p-extension. If $x \in D$, $a = v_D(x) > 0$, and $p \nmid a$, then for every $g \neq 1$,*

$$v_D(gx - x) = a + \ell_D(g). \tag{2.3}$$

Proof. The given coefficient field k is fixed by g, and $\mathcal{O}_D = k[[\pi]]$. The leading multiplier of $g\pi$ is a p-power root of unity in $k^\times$, hence equals 1. Write $g\pi = \pi(1 + u)$, with $v_D(u) = b = \ell_D(g) \geq 1$. For each positive integer j, characteristic p binomial expansion gives

$$v_D(g(\pi^j) - \pi^j) = j + p^{v_p(j)}b. \tag{2.4}$$

Indeed, writing $j = p^q j_0$, $p \nmid j_0$, reduces the bracket to $(1 + u^{p^q})^{j_0} - 1$, whose linear term is nonzero. In $x = c_a\pi^a + \sum_{j>a} c_j\pi^j$, the leading term has displacement order $a + b$, whereas every later term has order at least $j + b > a + b$. These orders tend to infinity, so the complete tail cannot cancel the leading term. $\square$

Applying this lemma to L_1/K, with $v_{L_1}(\alpha_1) = p - 1$, and using Lemma 2.3, proves that its unique break is $2(p - 1) - (p - 1) = p - 1$. More generally, (2.3) computes the first break of D/K as the minimum nonidentity x-displacement minus $v_D(x)$. Any nontrivial Galois quotient has first upper break at least this value, by upper quotient compatibility.

3 Exact interlevel contact and the second-layer degree

The first comparison supplies degree p^2 uniformly, without an Artin–Schreier coefficient calculation.

Proposition 3.1. *For every $e \geq 2$, $\alpha \in S_e$, and $\beta \in S_1$,*

$$v(\alpha - \beta) = d_0 := \frac{2(p-1)^2}{p^2}. \tag{3.1}$$

Moreover,

$$[K(\alpha) : K] \geq p^2, \qquad v(P^{\circ p}(\alpha) - \alpha) = 2(p - 1). \tag{3.2}$$

In particular, M_2 is irreducible of degree p^2.

Proof. The exact first factorization is

$$P^{\circ p}(z) - z = z(z+s)M_1(z)V_1(z). \tag{3.3}$$

Put $\mu = (P^{\circ p})'(\beta)$. Differentiating at β gives

$$\mu - 1 = \beta(\beta+s)M_1'(\beta)V_1(\beta).$$

The first two factors have valuation r, the last is a unit, and each of the $p-1$ root differences in $M_1'(\beta)$ has valuation $2r$. Hence

$$v(\mu-1) = 2r + (p-1)2r = 2(p-1) < +\infty. \tag{3.4}$$

For $e \geq 2$, both return polynomials defining Q_e vanish at β. Differentiate their polynomial quotient identity before evaluation. By the chain rule and (3.4), the derivative of the denominator is nonzero, and

$$Q_e(\beta) = \frac{\mu^{p^{e-1}} - 1}{\mu^{p^{e-2}} - 1} = (\mu-1)^{(p-1)p^{e-2}}.$$

This is not substitution into an undefined 0/0 expression. Since $V_e(\beta)$ is a unit, it follows that

$$\sum_{\alpha' \in S_e} v(\beta - \alpha') = 2(p-1)^2 p^{e-2}. \tag{3.5}$$

All contacts are finite, because the ordinary periods differ. Any two level-e points differ by a sum of one-step displacements and thus have difference valuation at least $2r$. The average in (3.5) is $d_0 < 2r$. Some contact is therefore below $2r$; the unequal-valuation triangle rule forces every contact to equal that one. Their average is d_0, proving (3.1).

Write $L = K(\alpha)$, $F = K(\beta) = L_1$, and $B = LF$. By Lemma 2.2, $[L:K] = p^h$, $1 \leq h \leq e$. Suppose $h = 1$. Since p is odd, d_0 has exact denominator p^2. The element $\eta = \alpha - \beta \in B$ therefore forces $[B:K] \geq p^2$; the opposite inequality follows from the two degrees. Thus $L \cap F = K$, $\mathrm{Gal}(B/K) \simeq C_p \times C_p$, and $v_B = p^2 v$. Because $h < e$, each nontrivial automorphism of L/K has a rotation index divisible by p, so moves α with valuation at least $3r$. Every nontrivial automorphism of F/K moves β with valuation $2r$. The restrictions can be chosen independently in the direct product. Hence the minimum nonidentity displacement of η has base valuation exactly $2r$.

But $v_B(\eta) = 2(p-1)^2$ is positive and prime to p. Lemma 2.4 makes the first break of B/K

$$p^2(2r - d_0) = 2(p-1).$$

Its quotient F/K has break $p-1$, contradicting upper quotient compatibility. Consequently $h \geq 2$. At $e = 2$, the orbit size also bounds the degree by p^2, so equality and irreducibility follow. Finally evaluate (3.3) at α. Both initial factors have valuation r, the complementary factor is a unit, and all p contacts equal d_0. Thus

$$v(P^{\circ p}(\alpha) - \alpha) = 2r + pd_0 = 2(p-1),$$

which proves (3.2). □

The contact (3.1) has denominator p^2 at every higher level. In particular, $p^e d_0$ is divisible by p when $e \geq 3$, so the prime-value argument cannot be repeated with this contact to force degree p^e. The next section uses level-two clusters instead.

4 Cluster transfer and full local inertia

For $j \geq 2$, define top clusters in S_j by

$$x \sim_j y \quad \Longleftrightarrow \quad x = y \text{ or } v(x-y) > 2r.$$

The ultrametric inequality makes this an equivalence relation.

Lemma 4.1. *There are exactly p top clusters in S_j. If $x \in S_j$, they are*

$$\mathcal{C}_{j,a} = \{P^{\circ(a+bp)}(x) : 0 \leq b < p^{j-1}\}, \qquad a \in \mathbb{Z}/p\mathbb{Z}.$$

Different clusters are separated by valuation $2r$, and P acts on the clusters by $a \mapsto a+1$. The Galois action on clusters is the reduction modulo p of its native rotation index.

Proof. Indices prime to p give displacement valuation $2r$, by Lemma 2.3. An index divisible by p gives a sum of p-step displacements. Each has valuation $2(p-1)$, by Proposition 3.1 and the isometry. Their sum has valuation at least $2(p-1) > 2r$. Thus two indices lie in the same cluster exactly when they agree modulo p. Galois preserves the root set and the valuation, and its rotation description is Lemma 2.2. $\square$

Fix $\beta \in S_2$, and let $\mu_2 = (P^{\circ p^2})'(\beta)$. Differentiating the exact identity

$$P^{\circ p^2}(z) - z = (P^{\circ p}(z) - z)M_2(z)V_2(z)$$

at β gives

$$\mu_2 - 1 = (P^{\circ p}(\beta) - \beta)M_2'(\beta)V_2(\beta).$$

There are $p(p-1)$ differences in the derivative product with rotation index prime to p, all of valuation $2r$. The remaining $p-1$ indices are ap, $1 \leq a < p$. Telescoping a p-step displacements, and dividing by the first, gives residue $a \neq 0$; hence these differences have valuation exactly $2(p-1)$. Consequently

$$v(\mu_2 - 1) = 2(p-1) + p(p-1)2r + (p-1)2(p-1) = 2(p-1)(2p-1). \tag{4.1}$$

In particular $\mu_2 \neq 1$.

Lemma 4.2 (Canonical matched clusters)**.** *For every $e \geq 3$, the p top clusters of S_e are canonically matched with those of S_2 by the relation*

$$C \longleftrightarrow D \quad \Longleftrightarrow \quad v(x-y) > 2r \text{ for some } x \in C,\ y \in D.$$

The matching is equivariant for both G_K and one application of P.

Proof. The derivative-ratio argument of Section 3, now at the ordinary period-p^2 point β, is legitimate by (4.1). It gives

$$Q_e(\beta) = \frac{\mu_2^{p^{e-2}} - 1}{\mu_2^{p^{e-3}} - 1} = (\mu_2 - 1)^{(p-1)p^{e-3}}.$$

The complementary factor is a unit. Therefore

$$\frac{1}{p^e} \sum_{\alpha \in S_e} v(\beta - \alpha) = \frac{2(p-1)^2(2p-1)}{p^3} > 2r.$$

The strict inequality follows from

$$\frac{2(p-1)^2(2p-1)}{p^3} - 2r = \frac{2(p-1)(p^2-3p+1)}{p^3} > 0$$

for every integer $p \geq 3$. Thus there is a pair $\alpha \in S_e, \beta \in S_2$ with contact deeper than $2r$. Applying $P^{\circ a}$, $0 \leq a < p$, gives p such pairs meeting all top clusters in both sets.

If one pair $x \in C, y \in D$ has contact deeper than $2r$, then every pair $x' \in C, y' \in D$ does: apply the ultrametric inequality to $x'-x$, $x-y$ and $y-y'$. If $C' \neq C$, its points have contact exactly $2r$ with points of C. The unequal-valuation triangle rule then gives contact exactly $2r$ between every point of C' and every point of D. Thus D cannot be matched to two clusters; the symmetric argument gives uniqueness for C. The p pairs already constructed prove existence everywhere. This matching is characterized by the valuation alone, hence is Galois-equivariant. Since P is an isometry, it carries a matched pair to a matched pair; uniqueness proves native equivariance as well. $\square$

Proof of the field assertion in Theorem 1.1. At level two, Proposition 3.1 gives $H_2 = C_{p^2}$, so G_K acts transitively on its p top clusters. Lemma 4.2 transfers that transitivity to the top clusters of S_e for every $e \geq 3$. If H_e were a proper subgroup of C_{p^e}, all its indices would be divisible by p; by Lemma 4.1, it would act trivially on those $p > 1$ clusters. Hence $H_e = C_{p^e}$. Level one is Lemma 2.2. The root action is transitive, giving irreducibility; the same lemma already proves that any root generates the totally ramified splitting field. Fullness of the native rotation subgroup supplies the designated generator $\alpha_e \mapsto P(\alpha_e)$. $\square$

5 Native-oriented Artin–Schreier classes

The field result permits a concrete, sign-fixed resolvent. For $n = p^e$ and $\alpha_e \in S_e$, set

$$w_e = \frac{\alpha_e^{n-1}}{M_e'(\alpha_e)}, \qquad y_e = -\sum_{i=0}^{n-1} \bar{i}\, \sigma_e^i(w_e), \qquad a_e = y_e^p - y_e, \tag{5.1}$$

where $\bar{i} \in \mathbb{F}_p$ is the residue of i.

Lemma 5.1 (Positive native translation)**.** *The elements in* (5.1) *satisfy*

$$\mathrm{Tr}_{L_e/K}(w_e) = 1, \qquad \sigma_e(y_e) - y_e = 1, \qquad a_e \in K, \qquad K(y_e) = F_e.$$

The class $[a_e] \in K/\wp(K)$ is independent of the starting root and of the trace-one choice, with this native orientation.

Proof. Let $x_0, \ldots, x_{n-1}$ be the distinct roots of M_e. The coefficient of T^{n-1} in Lagrange interpolation

$$T^{n-1} = \sum_{i=0}^{n-1} x_i^{n-1} \frac{M_e(T)}{(T-x_i)M_e'(x_i)}$$

gives $\sum_i x_i^{n-1}/M_e'(x_i) = 1$. This is the trace identity. Reindexing the weighted sum in (5.1), including its wrapped coefficient $-(n-1) = 1$ in characteristic p, gives $\sigma_e y_e - y_e = \sum_i \sigma_e^i w_e = 1$. Thus $\sigma_e^a y_e = y_e + \bar{a}$, a_e is invariant, and the stabilizer of y_e is precisely $\langle \sigma_e^p \rangle$. This proves $K(y_e) = F_e$.

Replacing w_e by any trace-one element produces y_e' with the same translation identity. Their difference is fixed by σ_e, hence lies in K, and $a_e' - a_e = \wp(y_e' - y_e)$. Replacing the starting root by $\sigma_e^a \alpha_e$ replaces y_e by $\sigma_e^a y_e = y_e + \bar{a}$, also preserving the class. No division by p^e has been used. $\square$

This is the additive trace-resolvent construction of classical Artin–Schreier theory, written here to specify the native sign. For completeness, every class in $K/\wp(K)$ has a unique reduced polar representative $\sum_{m>0,\ p\nmid m} c_m s^{-m}$ with finite support. The regular part is in $\wp(k[[s]])$: the constant equation is solvable in k, and the remaining coefficients are solved successively using the unit derivative -1. Subtracting $\wp(c^{1/p}s^{-m})$ eliminates a pole cs^{-pm}; starting from the largest pole terminates. A nonzero reduced polar polynomial cannot equal $\wp(b)$: if $v(b) < 0$ its largest pole would be divisible by p, and if $v(b) \geq 0$ it would have no pole. This also proves uniqueness.

Proposition 5.2 (Class stabilization). *For every $e \geq 2$, $[a_e] = [a_2]$ and $F_e = F_2$ inside the fixed separable closure.*

Proof. Let $\rho_e : G_K \to \mathbb{Z}/p^e\mathbb{Z}$ be characterized by

$$g\alpha_e = P^{\circ\rho_e(g)}(\alpha_e).$$

Commutation with P shows that the index is independent of the starting root; composition adds indices, so it is a continuous character, factoring through L_e/K. For $e \geq 3$, label the top clusters at each level by $i \in \mathbb{Z}/p\mathbb{Z}$, using native iteration. By Lemma 4.2, the matching is a map commuting with $i \mapsto i+1$, hence has the form $i \mapsto i+c$. Galois equivariance then gives, for every $g \in G_K$,

$$i + \overline{\rho_e(g)} + c = i + c + \overline{\rho_2(g)}.$$

Thus $\rho_e \bmod p = \rho_2 \bmod p$ on the whole group. The same assertion is tautological at $e = 2$. The translation identity in Lemma 5.1 gives

$$g(y_e) - y_e = \overline{\rho_e(g)} = \overline{\rho_2(g)} = g(y_2) - y_2.$$

Consequently $b_e = y_e - y_2$ is fixed by G_K and lies in K. It follows that $a_e - a_2 = \wp(b_e)$, and that $K(y_e) = K(y_2)$. The phase c permits only a translation of cluster labels, not a nontrivial scalar or sign change of the native generator. □

Proposition 5.3 (Separation of the prime-period field). *For every $e \geq 2$, $L_1 \cap L_e = K$.*

Proof. First suppose, for contradiction, $L_1 \subset L_2$. Choose $\alpha \in S_2, \beta \in S_1$, and put $\eta = \alpha - \beta \in L_2$. By Proposition 3.1,

$$v_{L_2}(\eta) = 2(p-1)^2, \qquad v_{L_2}(\sigma_2^p\eta - \eta) = 2p^2(p-1).$$

For the second equation, σ_2^p fixes the unique degree-p subfield, hypothetically L_1, and the p-step displacement of α is (3.2). The first value is prime to p, so Lemma 2.4 gives the second lower break, under this supposition, as

$$b_2 = 2p^2(p-1) - 2(p-1)^2 = 2(p-1)(p^2-p+1).$$

The degree-p quotient is hypothetically L_1/K, giving first break $b_1 = p-1$. But then

$$b_2 - b_1 = (p-1)(2p^2-2p+1) \equiv -1 \pmod{p},$$

contrary to (2.2). Thus $L_1 \not\subset L_2$, and its prime degree gives $L_1 \cap L_2 = K$.

If $L_1 \subset L_e$ at any later level, cyclicity would identify it with $F_e = F_2 \subset L_2$, contradicting what was just proved. Again prime degree gives the stated intersection. This completes Theorem 1.1. □

The second lower-break expression in this contradiction has not yet been proved to be an actual invariant: its paired first break arose from a false containment assumption. Section 6 derives the actual two breaks independently, with first break $2(p-1)$.

6 The first quotient and the second-layer different

We now calculate the actual ramification invariants. The element $\alpha_2 - \beta_1$ has valuation divisible by p in L_2L_1, so Lemma 2.4 does not apply there. Its replacement tracks the leading cancellation.

Lemma 6.1 (A once-p cancellation)**.** *Let D/K be a finite totally ramified Galois p-extension, with group G, first lower break $b_0 > 0$, and $p \nmid b_0$. Suppose $x \in D$ has $a = v_D(x) > 0$, $v_p(a) = 1$, and*

$$v_D(gx - x) > a + pb_0 \qquad (g \neq 1). \tag{6.1}$$

In any uniformizer expansion of x, the first nonzero exponent prime to p is

$$n = a + (p-1)b_0.$$

The first graded quotient G_{b_0}/G_{b_0+1} has order p. If g has later lower break $c = \ell_D(g) > b_0$, then

$$v_D(gx - x) = n + c. \tag{6.2}$$

Proof. Write $x = \sum_{j \geq a} c_j \pi^j$, with $c_a \neq 0$. For an element g of break b_0, (2.4) says that the leading monomial contributes at exact order $a + pb_0$. Every later monomial with exponent divisible by p contributes at strictly greater order. If the first nonzero exponent j prime to p were less than $a + (p-1)b_0$, its contribution at $j + b_0$ would be uniquely lowest, contradicting (6.1). If no such exponent were present at $n = a + (p-1)b_0$, the leading monomial at order $a + pb_0$ could not cancel. Thus n is the first such exponent, and $p \nmid n$.

To control the entire first grade, not just one automorphism, put

$$\theta(g) = \operatorname{res}\left(\frac{g\pi - \pi}{\pi^{b_0+1}}\right), \qquad g \in G = G_{b_0}.$$

Composition adds the leading coefficients, since each automorphism fixes k and has linear coefficient 1. Hence θ is additive and has kernel G_{b_0+1}. The coefficient of order $a + pb_0$ in $gx - x$ is

$$c_a\,\overline{(a/p)}\,\theta(g)^p + c_n \overline{n}\,\theta(g).$$

Only these two monomials contribute at that order. By (6.1), the displayed expression vanishes for every g. Its leading coefficient is nonzero, so this degree-p polynomial has at most p roots. The image of the nontrivial first graded p-group therefore has exactly p elements.

For a later break $c > b_0$, the monomial $c_n\pi^n$ contributes at order $n + c$. Later prime-to-p exponents give larger orders. Every exponent divisible by p gives order at least $a + pc$, and

$$a + pc - (n + c) = (p-1)(c - b_0) > 0.$$

The contribution at $n + c$ is unique. All tail orders tend to infinity, so no infinite-series cancellation changes it. This proves (6.2). $\square$

6.1 Apply the cancellation in the disjoint compositum

Set $B = L_2L_1$, $G = \mathrm{Gal}(B/K)$, and choose $\alpha \in S_2, \beta \in S_1$. By Proposition 5.3,

$$G \simeq C_{p^2} \times C_p, \qquad v_B = p^3 v.$$

Let $\sigma, \gamma \in G$ act by one native step on α, β, respectively, fixing the other root. For $\eta = \alpha - \beta$, write

$$a = v_B(\eta) = 2p(p-1)^2, \qquad A = 2p^2(p-1). \tag{6.3}$$

The contact gives the first equality, with $v_p(a) = 1$. Both root displacements have base valuation at least $2r$, so

$$v_B(g\eta - \eta) \geq A \quad (g \neq 1), \qquad v_B(\sigma\eta - \eta) = v_B(\gamma\eta - \eta) = A. \tag{6.4}$$

Infinite valuation is allowed in the inequality.

Let b_0 be the first lower, equivalently upper, break of B/K. Its quotient L_1/K gives $b_0 \leq p - 1$. The first drop of a finite abelian p-extension is detected by a degree-p character: the first graded quotient is elementary abelian (the leading-coefficient injection above shows this), and a nonzero character of that quotient pulls back to such a character of G. Its Artin–Schreier break is b_0. Thus b_0 is positive and prime to p. Finally

$$A - a = 2p(p-1) > p(p-1) \geq pb_0.$$

Lemma 6.1 applies. In particular the first graded quotient has order p, and with

$$n = a + (p-1)b_0 \tag{6.5}$$

we have $v_B(g\eta - \eta) = n + \ell_B(g)$ whenever $\ell_B(g) > b_0$.

6.2 Exclude all early-conductor orderings

Put $F = F_2$, let b be its unique break, and write u_2 for the second upper break of L_2/K. Classical cyclic ramification gives $u_2 \geq pb \geq p$. Let

$$E = FL_1, \qquad N = \mathrm{Gal}(B/E) = \langle \sigma^p \rangle.$$

Then E/K is elementary abelian of degree p^2. The subgroup N persists in G^u for $0 \leq u \leq p-1$. Indeed choose $p - 1 < u' < u_2$. The projection of $G^{u'}$ on L_1 is trivial, while its projection on L_2 contains $\langle \sigma^p \rangle$. In the direct product these two projections force all of N to lie in $G^{u'}$; monotonicity gives the assertion. It follows that

$$[G : G^u] = [G/N : (G/N)^u] \qquad (0 \leq u \leq p-1). \tag{6.6}$$

This is a quotient argument, not an assumed product rule for compositum ramification.

Suppose $b \leq p - 1$. All degree-p characters of E/K are linear combinations over $\mathbb{F}_p$ of those for F/K and L_1/K. Their reduced polar representatives have largest poles at most $p - 1$; a linear combination cannot introduce a larger pole. The L_1 character has break exactly $p - 1$. Characters separate this elementary abelian group, so its last upper break is $p-1$. By (6.6) and the cancellation lemma its first graded quotient has order p. Hence E/K has exactly two distinct upper breaks, $b_0 < p - 1$ and $p - 1$, each dropping the group by a factor p. This includes the possibility $b = p - 1$: possible cancellation of the two leading poles either produces a smaller first break, or gives a single drop of order p^2, which the lemma rules out.

The lower value corresponding to the last break of E/K is

$$c = \psi_B(p-1) = b_0 + p(p - 1 - b_0) > b_0.$$

There is $h \in G$ with lower break exactly c: lift a nontrivial element of the last upper group of E/K inside G^{p-1}; its image disappears immediately after that break. By (6.5) and (6.2),

$$v_B(h\eta - \eta) = n + c = a + p(p-1) < A,$$

contradicting (6.4). Therefore

$$b > p - 1. \tag{6.7}$$

6.3 Read off the exact first quotient break

By (6.7), every nonzero linear combination involving the character of F/K has break b: its highest pole cannot cancel with the strictly lower pole of the L_1 character. The other nonzero characters have break $p-1$. Thus E/K has upper breaks $p-1, b$, each of rank one. The subgroup N persists through upper b: choose $b < u' < u_2$, possible since $u_2 \geq pb > b$, and repeat the two-projection argument. Consequently

$$b_0 = p-1, \qquad n = a + (p-1)^2.$$

For $p-1 < u \leq b$, the L_1 projection is trivial and the L_2 projection is full. Hence $G^u = \langle \sigma \rangle$, and immediately after b its L_2 projection drops to $\langle \sigma^p \rangle$. The lower break of the native generator σ is thus

$$\ell_B(\sigma) = \psi_B(b) = (p-1) + p(b-(p-1)) > b_0.$$

Its displacement is A; the later-break rule yields

$$\ell_B(\sigma) = A - n = 2p^2(p-1) - 2p(p-1)^2 - (p-1)^2 = p^2 - 1.$$

Comparison gives

$$(p-1) + p(b-(p-1)) = p^2 - 1, \qquad b = 2(p-1). \tag{6.8}$$

Since $F_e = F_2$ for $e \geq 2$, this proves the asserted first quotient break at every higher level. Equivalently, the largest pole of the native-oriented reduced AS class is $2(p-1)$, with nonzero coefficient. No general formula for all its coefficients has been used or obtained.

6.4 The complete second-layer filtration

The full upper filtration of B/K, at this point with the last break u_2 still undetermined, is

$$G^u = \begin{cases} G, & 0 \leq u \leq p-1, \\ \langle \sigma \rangle, & p-1 < u \leq 2(p-1), \\ \langle \sigma^p \rangle, & 2(p-1) < u \leq u_2, \\ 1, & u > u_2. \end{cases} \tag{6.9}$$

The first two lines were proved above. After $p-1$ the L_1 projection is trivial, and upper quotient compatibility identifies the L_2 projection with its upper group. A subgroup of $C_{p^2} \times C_p$ with trivial second projection is uniquely determined by its first projection. This proves the last two lines without any additional independence assumption.

Let c_0, c_1, c_2 be the lower breaks of B/K. The first two are

$$c_0 = p-1, \qquad c_1 = (p-1) + p(p-1) = p^2 - 1.$$

The element $\tau = \sigma^p$ has lower break c_2. It fixes β, so (3.2) gives

$$v_B(\tau\eta - \eta) = 2p^3(p-1).$$

The later-break rule is applicable because $c_2 > c_1 > b_0$. Subtracting its identity for σ from that for τ gives

$$c_2 - c_1 = 2p^3(p-1) - 2p^2(p-1) = 2p^2(p-1)^2.$$

The index on the last interval of (6.9) is p^2. Hence Herbrand's formula gives

$$u_2 - 2(p-1) = \frac{c_2 - c_1}{p^2} = 2(p-1)^2, \qquad u_2 = 2p(p-1).$$

In particular, the upper and lower breaks of B/K are

$$\begin{aligned}&(p-1,\ 2(p-1),\ 2p(p-1)),\\&(p-1,\ p^2-1,\ p^2-1+2p^2(p-1)^2).\end{aligned}\tag{6.10}$$

Upper quotient compatibility identifies the two upper breaks of L_2/K. Its lower breaks are

$$b_1 = 2(p-1), \qquad b_2 = b_1 + p\big(2p(p-1) - 2(p-1)\big) = 2(p-1)(p^2-p+1).$$

These are actual invariants, derived with the actual first break, independently of the hypothetical calculation in Proposition 5.3.

6.5 Different exponents versus the root order

For a finite totally ramified Galois extension D/K, the Hilbert different formula in the integer-normalized valuation is

$$d_{D/K} = \sum_{i\geq 0}(|\operatorname{Gal}(D/K)_i| - 1). \tag{6.11}$$

One way to check this normalization is to use a uniformizer π, which generates the integer ring over R. Its minimal polynomial is Eisenstein; its derivative generates the different. The valuation of that derivative is $\sum_{g\neq 1} v_D(g\pi - \pi)$, and counting each automorphism in its lower groups gives (6.11). The derivative description is the monogenic case of the classical different formula [9, Tags 0BWD and 0BWG].

For L_2/K, the successive lower group orders are p^2, p, giving

$$\begin{aligned}d_{L_2/K} &= (b_1+1)(p^2-1) + (b_2-b_1)(p-1)\\&= (2p-1)(p^2-1) + 2p(p-1)^3\\&= (p-1)(2p^3-2p^2+3p-1).\end{aligned}$$

This proves Theorem 1.2. For B/K, the three lower group orders are p^3, p^2, p, so the additional invariant is

$$\begin{aligned}d_{B/K} &= (c_0+1)(p^3-1) + (c_1-c_0)(p^2-1) + (c_2-c_1)(p-1)\\&= p(p^3-1) + p(p-1)(p^2-1) + 2p^2(p-1)^3\\&= p^2(p-1)(2p^2-2p+3).\end{aligned}$$

These are field different exponents, not the discriminant of $R[\alpha_2]$. For example, at $p=3$ the proved formulas give upper breaks $(4,12)$, lower breaks $(4,28)$, and $d_{L_2/K} = 88$. The root-difference product instead gives

$$v(\operatorname{Disc}(M_2)) = p^2\big(p(p-1)2r + (p-1)2(p-1)\big) = 4p^2(p-1)^2,$$

which equals 144 at $p=3$. This is a direct consequence of the distances already proved, not a numerical input to the field calculation. It must not be substituted for the exponent 88.

7 The eventual native quotient tower

This section combines the local Galois fields with an independent compact-limit theorem. We state the imported interface explicitly before proving the Galois continuity and quotient transfer needed for Theorem 1.3.

Choose an algebraic closure of K containing K^{sep}, and let $\mathcal{C}$ be its completion. It is complete and algebraically closed by the classical completion theorem [2, Theorem 1.1]. Fix $0 < |s| < 1$ and use the associated absolute value. The sets S_e are still the unique small ordinary cycles in $\mathcal{C}$, by Lemma 2.1.

Theorem 7.1 (Compact adding-machine input)**.** *For these cycles, the ordinary metric closure*

$$C = \overline{\bigcup_{e\geq 1} S_e}^{\mathcal{C}}$$

is compact. The set

$$\Omega = \bigcap_{N\geq 1} \overline{\bigcup_{e\geq N} S_e}^{C} \tag{7.1}$$

is nonempty and compact, $P(\Omega) = \Omega$, and $d_H(S_e, \Omega) \to 0$ for the full sequence. There is a homeomorphism $h : \Omega \to \mathbb{Z}_p$ such that

$$h(Px) = h(x) + 1. \tag{7.2}$$

This is precisely the compactness, Hausdorff and conjugacy part of the unpublished companion [1, Theorem 1.1, *Compact adding-machine limit of the optimal cycles*], applied to $\mathcal{C}$ and $\lambda = 1 + s$. The companion proves it for every complete algebraically closed nonarchimedean field of characteristic p and every $0 < |\lambda - 1| < 1$. Its proof includes the all-anchor identity

$$\frac{1}{p^e} \sum_{\alpha\in S_e} v(\beta - \alpha) = c_d := \frac{p-1}{p^{d+1}} v\big((P^{\circ p^d})'(\beta) - 1\big) \geq dr^3 + r^2(1 + 1/p), \qquad e > d, \quad \beta \in S_d.$$

The finite value of this average excludes the finite-cycle alternative for the compact limit. Thus no unproved isolation assumption, nor merely weak convergence of measures, is being used in (7.2). The companion proves its factor, contact and compactness assertions without using full local inertia or the first AS character. Theorem 7.1 is the only additional model-specific input in this section; no part of the preceding sections depends on it.

7.1 Continuity of the Galois action on the compact closure

The characters $\rho_e : G_K \to \mathbb{Z}/p^e\mathbb{Z}$ were defined in Proposition 5.2. By Theorem 1.1 they are continuous and surjective, with one native step labeled $+1$. Every $g \in G_K$ extends uniquely to the algebraic closure: purely inseparable roots have uniquely determined images. The extension of the valuation from complete K is unique, so the resulting action is isometric. It therefore extends to an isometric automorphism of $\mathcal{C}$, with inverse the extension of g^{-1}. Since $gS_e = S_e$ at every level and g commutes with P, it preserves C, every tail closure in (7.1), and Ω.

We need continuity in g, not just individual isometries. Fix $\varepsilon > 0$. Compactness of C and density of the algebraic cycle union give a finite set $A_\varepsilon \subset \bigcup_e S_e$ such that every $x \in C$ is within distance strictly less than ε of some $a \in A_\varepsilon$. Each a is separable algebraic over K, so its stabilizer in G_K is open. Let U_ε be the intersection of these finitely many stabilizers. For $g \in U_\varepsilon$,

$$|gx - x| \leq \max\{|gx - ga|, |ga - a|, |a - x|\} < \varepsilon \qquad (x \in C).$$

Thus elements of G_K near the identity move all of C uniformly little. If $g = g_0u$, $u \in U_\varepsilon$, and $|x - x_0| < \varepsilon$, then

$$|gx - g_0x_0| = |ux - x_0| \leq \max\{|ux - x|, |x - x_0|\} < \varepsilon.$$

This proves joint continuity of $G_K \times C \to C$, and of its restriction to Ω. It requires no assertion that a limit point is algebraic.

7.2 A canonical translation character

For a homeomorphism h satisfying (7.2), the continuous map $f_g = h \circ g \circ h^{-1} : \mathbb{Z}_p \to \mathbb{Z}_p$ commutes with addition by 1. Any continuous f with $f(z+1) = f(z) + 1$ is a translation: induction gives $f(m) = f(0) + m$ on nonnegative integers, and these integers are dense in $\mathbb{Z}_p$. Hence there is a unique $\chi_\infty(g) \in \mathbb{Z}_p$ such that

$$h(gx) = h(x) + \chi_\infty(g) \qquad (x \in \Omega). \tag{7.3}$$

Composition adds translation amounts, so χ_∞ is a homomorphism. For a fixed $\omega_0 \in \Omega$,

$$\chi_\infty(g) = h(g\omega_0) - h(\omega_0),$$

and the joint continuity just proved shows that it is continuous.

The character is independent of the origin and of the chosen native conjugacy: if h' also satisfies (7.2), then $h' \circ h^{-1}$ commutes with addition by 1, so is itself a translation. Conjugating a translation by a translation does not change its amount. Multiplication of coordinates by a unit $u \in \mathbb{Z}_p^\times$ would instead change the native step to $+u$, and is compatible with (7.2) only if $u = 1$. There is therefore no sign or scalar ambiguity at any finite depth.

7.3 Transfer every finite quotient to all late cycles

Fix $j \geq 1$ and define the continuous surjection

$$q_j : \Omega \longrightarrow \mathbb{Z}/p^j\mathbb{Z}, \qquad q_j(x) = h(x) \bmod p^j.$$

Its finitely many fibers are compact and clopen. They are permuted natively by $+1$ and by Galois translation by $\chi_\infty(g) \bmod p^j$. This quotient exists at every j; there is no assumption that a particular family of metric partitions attains every cardinality p^j. Equivalently, a cofinal inverse system of cyclic partitions suffices, since any partition of size p^k, $k \geq j$, maps to the quotient modulo p^j.

Different fibers of q_j have positive minimum distance: the distance function attains its minimum on their compact product, and disjointness makes that minimum nonzero. Choose $\delta_j > 0$ such that

$$x, y \in \Omega,\ |x - y| < \delta_j \quad \Longrightarrow \quad q_j(x) = q_j(y). \tag{7.4}$$

Full-sequence Hausdorff convergence supplies $E_j \geq j$ with

$$d_H(S_e, \Omega) < \delta_j \qquad (e \geq E_j). \tag{7.5}$$

This threshold is chosen before, and independently of, g.

For each such e, define $q_{e,j} : S_e \to \mathbb{Z}/p^j\mathbb{Z}$ by choosing $x \in \Omega$ with $|\alpha - x| < \delta_j$ and setting $q_{e,j}(\alpha) = q_j(x)$. Two choices of x are at mutual distance less than δ_j, by the ultrametric inequality; hence (7.4) makes the definition independent of the choice. The other directed Hausdorff bound in (7.5) proves surjectivity.

All points in C lie on $|z| = |s|^r < 1$, by continuity of the norm and Lemma 2.1. On the open unit disk,

$$P(x) - P(y) = (x - y)(1 + s + x + y), \qquad |1 + s + x + y| = 1.$$

Thus P is an isometry there. If x is an allowed choice for α, then Px is an allowed choice for $P\alpha$. Likewise gx is an allowed choice for $g\alpha$, since Galois acts isometrically and preserves Ω. Therefore

$$\begin{aligned} q_{e,j}(P\alpha) &= q_{e,j}(\alpha) + 1, \\ q_{e,j}(g\alpha) &= q_{e,j}(\alpha) + \chi_\infty(g) \bmod p^j. \end{aligned} \tag{7.6}$$

There is no need to choose nearby points globally consistently; choice independence was proved first. On the other hand, the definition of ρ_e and the first identity give

$$q_{e,j}(g\alpha) = q_{e,j}(P^{\circ \rho_e(g)}\alpha) = q_{e,j}(\alpha) + \rho_e(g) \bmod p^j.$$

Comparison with the second identity in (7.6) proves (1.4) simultaneously for every $g \in G_K$. It is exact equality of characters, not just an abstract isomorphism of finite cyclic fields or a subsequence limit.

7.4 Surjectivity and the algebraic kernel fields

The image $H = \chi_\infty(G_K)$ is compact and hence closed in $\mathbb{Z}_p$, since G_K is profinite. Taking $j = 1$ and a sufficiently late e in (1.4) shows that its reduction modulo p is the nonzero native first character. Thus H contains a p-adic unit u. It contains all integer multiples of u, which are dense in $\mathbb{Z}_p$, and closedness yields $H = \mathbb{Z}_p$.

For every j, $\chi_\infty \bmod p^j$ is a continuous surjection onto the cyclic group of order p^j. Its kernel is open and normal, so Galois correspondence [8, Section 9.22] gives

$$K_j = (K^{\mathrm{sep}})^{\ker(\chi_\infty \bmod p^j)}, \qquad [K_j : K] = p^j.$$

For $e \geq j$, the unique degree-p^j subfield of L_e is

$$F_{e,j} = L_e^{\langle \sigma_e^{p^j} \rangle} = (K^{\mathrm{sep}})^{\ker(\rho_e \bmod p^j)}.$$

The proved character equality therefore gives

$$K_j = F_{e,j} \qquad (e \geq E_j)$$

as actual subfields of the chosen separable closure. The kernels for $j + 1$ lie in the kernels for j, so $K_j \subset K_{j+1}$. Restrictions on Galois groups are the ordinary reductions modulo p^j, induced by the same character. Consequently

$$\mathrm{Gal}\left(\bigcup_{j \geq 1} K_j \,/\, K \right) \simeq \varprojlim_j \mathbb{Z}/p^j\mathbb{Z} = \mathbb{Z}_p,$$

with quotient map χ_∞. If a second continuous character satisfied (1.4), choose e beyond both thresholds at each j. Their reductions both equal $\rho_e \bmod p^j$; equality at every j gives equality in $\mathbb{Z}_p$. This proves uniqueness and completes Theorem 1.3.

In particular $K_1 = F_2$, by choosing e beyond E_1 and using Proposition 5.2. It is not L_1, since $L_1 \cap L_2 = K$. For larger j, the theorem fixes the quotient degree first and only then takes the cycle level sufficiently large. It does not imply containment of a prescribed full L_j in all later fields.

Finally, the points of Ω are classical elements of the completed algebraic closure, not necessarily algebraic over K. No finite Galois correspondence has been applied to such a coordinate, and no theorem identifying fixed fields inside the completion is used. Every field above is defined by a continuous character kernel inside K^{sep}.

8 Scope and further questions

The local arithmetic separates three objects attached to the same native cycles. The full splitting field L_e has degree p^e; its first quotient F_e equals F_2 for $e \geq 2$; and for a fixed j the eventual kernel field K_j is the common degree-p^j quotient of all sufficiently late L_e. Only the last collection has a proved nesting relation. The prime-period field L_1 is separated from every higher L_e, while the common first quotient has twice its ramification break.

The exact higher ramification sequence of L_e/K, general-prime coefficients of the reduced first AS representative, effective thresholds E_j, and complete pairwise intersections of the full L_e are not determined here. Nor does a theorem about this completed small factor decide transitivity on all global dynatomic cycles. These are distinct questions, rather than consequences of the cyclicity or of the compact-limit construction.

Preparation and proof provenance

This manuscript was prepared with AI assistance and current-team internal proof checking. These activities are not external peer review or publication acceptance. The arithmetic contact, native cluster-transfer and ramification arguments form one integrated local-field result. The independently proved compact-limit theorem is explicitly credited to the unpublished companion [1], and is used only in Section 7. All remaining new arguments used for the conclusions are given in this article. No numerical experiment or computer-assisted mathematical certificate is required by the proofs.

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